

PROJECT PLAN | OWNER: PM (KYC) | ~10 WKS BUILD, ~12 WKS RAMP | FOR APPROVAL

KYC Verification Waterfall v2 - Project Plan

We ramp at the speed of the queue. Ten workstreams over twenty-two weeks, ~8 additional review FTE, and a five-phase rollout - the critical path runs through analyst hiring, not code.

This document is self-contained. It states the problem, the fix, what it costs, who does what, in what order, and how the rollout limits risk. The **PRD** holds the design and requirements; the **ARD / Monitoring Plan** holds metric definitions. Neither is needed to read this.

1. The ask, in one page

The problem. Every applicant for our US consumer product must pass identity verification (KYC). Verification runs as a waterfall — a primary provider, then secondary providers, then a human reviewer. **7.93% of applicants end up not verified, against a 5% bar.** Across 2,515,155 applications (Jan 2023 – May 2026) that is 199,421 people, roughly **73,663 more than the bar allows**.

The cause is not that people fail checks — the waterfall stops running. 101,162 applicants (half of all non-verifications) failed the primary check and then had **no second check of any kind**. Nobody looked at them again. Comparable applicants who *were* routed onward recovered at **48%**. Three quarters of all non-verification is process failure, not a correct decision.

The fix. Rebuild the routing so every failure has a defined path and no applicant can leave the system undecided. **No verification standard, threshold, or control is loosened** — the entire improvement comes from making the waterfall finish. Two changes actually *strengthen* controls: every applicant now gets the primary check (today 3.74% bypass it), and sanctions/PEP screening becomes a distinct, fast-routed signal instead of being folded into an ordinary failure.

What it buys, what it costs.

Expected outcome	Non-verification 7.93% → ~6.0% ; committed target ≤6.3% , re-baselined on live data at 50% rollout
Principal cost	Human review volume rises ~2.4x (~21,900 → ~51,600 cases/yr) — ≈8 additional analyst FTE
Provider spend	Roughly +\$120K–240K/yr in additional secondary checks (<i>illustrative unit costs on real volumes; contracted rates pending</i>)
Timeline	~10 weeks to build · ~12 weeks to ramp · ~22 weeks end to end
Critical path	Analyst hiring, not code. Engineering is ramp-ready around week 13; review capacity is not ready for 100% until roughly week 22.

The honest headline for this committee: *we are proposing to recover roughly 49,000 wrongly-lost applicants a cycle by asking human reviewers to do about 2.4x the work.* That is a staffing decision as much as an engineering one, and the plan is built so that the ramp moves at the speed of the queue, not the speed of the build.

2. Objectives and success criteria

#	Objective	Today	Target
1	No applicant leaves the waterfall undecided	101,162 stranded	0
2	Reduce non-verification	7.93%	≤6.3% (5.0% stretch)
3	Every applicant gets the primary check	3.74% bypass it	100%

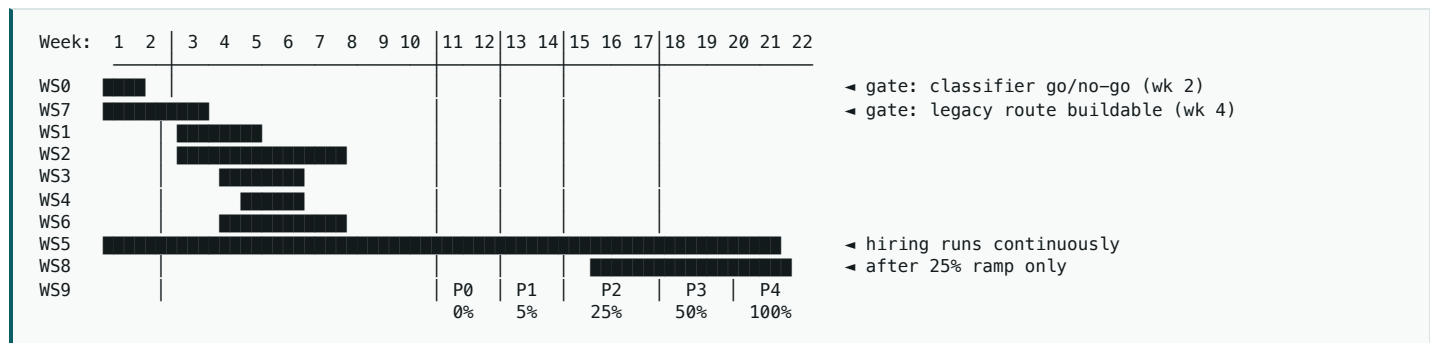
#	Objective	Today	Target
4	A clean pass is final	5.23% got extra checks anyway	0
5	The human safety net actually catches	reaches 30.2% of eligible cases	100%
6	Sanctions/PEP routed to compliance fast	no distinct route	100%, within 4h
7	No control weakened to buy the rate	—	Fraud, decline-rate and sanctions guardrails all within tolerance

Objective 3 will cost us rate, and we are doing it anyway. Applicants who bypass the primary check today verify at 99.08%. Forcing them through a real check will fail some of them. It is measured on its own line so it is never confused with the routing improvement — and it is a control fix, not a conversion play.

3. Workstreams

WS	Workstream	Owner	Depends on	Deliverable
WS0	Discovery & vendor confirmation	PM + Backend Eng Lead	—	A signed answer to: <i>does our primary provider return structured failure-reason codes and a separable sanctions flag?</i> Plus contracted provider unit costs and latency SLAs. Go/no-go on the classifier design.
WS1	Reason classifier	Backend Eng	WS0	Deterministic rule layer mapping provider responses to one of four reason classes, plus the secondary-issue signal
WS2	Orchestration engine	Backend Eng	WS1 (interface only)	Routing tree, mandatory primary check, unconditional pass-exit, guaranteed terminal state, retry/timeout handling, per-change feature flags
WS3	Provider integration cleanup	Integrations Eng	WS0	Duplicate LexisNexis integrations unified into one secondary role; Persona SSN and IDV modes exposed as distinct tiers; the undocumented primary entry point removed
WS4	Sanctions/PEP split & compliance path	Compliance Ops Lead + Backend	WS1	Independent flag, immediate compliance route, compliance queue separated from identity review, audit trail
WS5	Review capacity & tooling	Review Ops Manager	WS0 volume model	Phased hiring plan (≈8 FTE), case UI showing reason class and full path, SLA definitions, escalation policy
WS6	Data & monitoring	Data & Analytics	WS2 event schema	Decision-event table, five dashboards, alert rules, daily reconciliation job
WS7	"Legacy account" definition	PM + Account Services	—	An operational rule for the legacy-SSN route. Currently undefined — no signal in our data (age, vintage, or partner) reproduces it
WS8	Stranded backlog migration	Review Ops + Backend + Legal	WS2, WS5, Legal sign-off	Throttled re-processing of the 101,162 existing stranded applicants, with notification copy cleared by Legal
WS9	Rollout & measurement	PM + Data	WS6	Cohort assignment, phase gates, rollout scorecard, go/no-go decisions

4. Sequencing



Critical path: WS0 → WS1 → WS2 → WS6 → shadow → ramp. **Parallel, off the critical path:** WS3, WS4, WS7. **Paces the second half:** WS5. Engineering is ready to ramp around week 13; analyst capacity is not ready for 100% traffic until roughly week 22. **We ramp at the speed of the queue.**

5. Dependencies, and what breaks if they slip

Dependency	Owner	Risk if unresolved	Mitigation
Primary provider returns structured reason codes	WS0	The classifier — and with it the whole routing design — cannot be built as specified. This is the single load-bearing assumption in the programme and it is unconfirmed .	A fallback design is already specified: on any failure, run the SSN-path check and the secondary identity check in parallel, then document verification, then human review. It still delivers objectives 1, 3, 4 and 5; it costs more per failed applicant and loses the sanctions fast-path. Decide by end of week 2 — do not start WS1 on an assumption.
"Legacy account" has a definition	WS7	The legacy SSN route has no evaluable gate	If unresolved by week 4, ship v1 routing all SSN mismatches to LexisNexis, logged as a documented, accepted deviation. Add the legacy route when Account Services can define it.
≈8 analyst FTE hired and trained	WS5	Ramping past 25% creates a review backlog — a worse applicant experience than today's stranding	Ramp gates are tied to headcount, not sprint completion. Contract/BPO capacity as a bridge if hiring lags.
Legal sign-off on backlog notification	WS8	The 101,162 stranded applicants cannot be re-processed	WS8 is sequenced last by design so it never blocks the main rollout.
Contracted provider unit costs	WS0	Every cost figure in this plan stays illustrative	Contract data in by week 2; cost thresholds re-set before the 25% ramp.

6. Rollout, and how it limits risk

Phase	Traffic	Duration	Gate to enter	Gate to exit
P0 Shadow	100% mirrored, 0% acting	2 wks	Built, dashboards live	Classifier $\geq 95\%$ agreement with human-labelled cases; projected review volume within $\pm 20\%$ of model
P1 Canary	5%	2 wks	+2 analysts onboarded	Zero stranded applicants in cohort; no critical alerts; review turnaround p90 <72h
P2 Ramp	25%	3 wks	+3 analysts	Non-verification in cohort $\leq 6.5\%$; document-verification completion $\geq 65\%$; guardrails flat
P3 Majority	50%	3 wks	+3 analysts; queue turnaround stable	Guardrails within tolerance; target re-baselined on live data
P4 Full	100%	—	All gates green	Legacy path deleted after 30 days dark

Five deliberate risk controls:

1. **Shadow before acting.** P0 runs the new system on all traffic while making zero real decisions — the only phase where a wrong classifier costs nothing. Given the unconfirmed vendor dependency, skipping shadow would be the worst decision available to us.
2. **One switch per change.** Routing, pass-exit, bypass retirement and document-verification promotion each ship behind their own flag. Bypass retirement is *expected* to cost verification rate; bundled with the routing fix it would mask a genuine win and we would draw the wrong conclusion about the whole programme.
3. **Halt, don't auto-rollback.** An unresolved critical alert freezes the ramp at its current percentage. It does not automatically revert — flipping routing mid-application is its own failure mode. Rollback is a human decision, with the old path kept warm through P4.
4. **A control group survives to 100%.** The metric that ultimately decides whether this was a good idea is post-verification fraud rate, and it lags 90 days. Cutting over early destroys the comparison and leaves the question permanently unanswerable.
5. **The backlog migration is throttled and reported separately.** Re-processing the 101,162 stranded applicants is capped at 20% of daily review capacity, and its recoveries are excluded from the steady-state rate so the running system is not flattered by a one-off.

7. Cost and resourcing

Item	Basis	Estimate
Additional human review	~29,700 extra cases/yr at ~\$8/case	≈ +\$238K/yr
Additional review headcount	15 cases/analyst/day	≈ 8 FTE , hired in three tranches gated to the ramp
Additional provider calls	~100K–160K extra secondary checks/yr at a blended ~\$1.00–1.50	≈ +\$120K–240K/yr
Engineering	2 backend, 1 integrations, 1 analytics engineer	~10 weeks

Item	Basis	Estimate
Ongoing run-rate increase		≈ \$360K–480K/yr

All unit costs above are illustrative; the volumes are real and measured. Contracted vendor rates and true per-case ops costs land in week 2 (WS0) and every figure here is restated before the 25% ramp. No budget commitment should be made against these numbers until then.

The structural answer to this cost is out of v1 scope, on purpose. A risk-based auto-decline layer — scoring which automated failures are safe to reject without a human look — is the lever that would cut most of the 2.4× review increase. It needs labelled outcome data that this build will itself generate, which is why it is scoped as the v2.1 fast-follow rather than guessed at now.

8. Top risks

Risk	Likelihood	Impact	Owner	Response
Classifier not buildable as designed	Medium	High	WS0	Fallback design pre-specified; decision forced at week 2
Review queue overruns	Medium	High	WS5	Capacity-gated ramp; BPO bridge; auto-decline layer as the structural fix (v2.1)
Document-verification abandonment at 10–20× volume	Medium	Medium	PM	Invitation / start / completion instrumented separately; the step is placed last in the path; ships on its own flag
Recovery lands below the modelled 48%	Medium	Medium	PM + Data	Target re-baselined at 50% rollout on live data — stated up front, not discovered late
Bypass retirement measurably lowers the rate	High	Low–Med	PM	Expected. Isolated on its own flag and reported on its own line. It is a control fix; we ship it and report it honestly rather than trading it for headline rate.
A control weakens invisibly	Low	Severe	Compliance	Three guardrail metrics; sanctions guardrail paged as critical; control group held to 100%

9. Decisions needed from this group

#	Decision	Needed by
1	Approve ≈8 additional review FTE and the ~\$360–480K/yr run-rate increase, phased against the ramp	Week 1 — hiring is the critical path
2	Accept ≤ 6.3% as the committed target, with re-baselining at 50% rollout, rather than committing to 5.0% on day one	Week 1
3	Account Services to define "legacy account", or accept shipping without that route as a documented deviation	Week 4

#	Decision	Needed by
4	Legal/Compliance to approve re-contacting the 101,162 stranded applicants	Before week 15
5	Accept that the primary-check bypass retirement will likely <i>reduce</i> headline verification rate, as the price of closing a control gap	Week 1

10. Definition of done

- Zero stranded applicants, zero checks after a clean pass, and 100% terminal-state completeness held for **30 consecutive days at full traffic**.
- Non-verification rate re-baselined and reported against the 5% bar, with the treatment/control comparison shown.
- Decline-rate and sanctions guardrails within tolerance; the first 90-day post-verification fraud cohort read out.
- Stranded backlog fully re-processed; legacy routing path deleted.
- The "legacy account" question either implemented or closed as an accepted, documented deviation.
- Auto-decline layer scoped for v2.1 using the outcome data this build has by then generated — the lever that pays back the review cost this design deliberately took on.